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Material recovery from electronic waste using pyrolysis: Emissions measurements and risk assessment

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Abstract

Electronic waste (e-waste) generation has been growing in volume worldwide, and the diversity of its material composition is increasing. Sustainable management of this material is critical to achieving a circular-economy and minimizing environmental and public health risks. This study's objective was to investigate the use of pyrolysis as a possible technique to recover valuable materials and energy from different components of e-waste as an alternative approach for limiting their disposal to landfills. The study includes investigating the potential environmental impact of thermal processing of e-waste. The mass loss and change in e-waste chemicals during pyrolysis were also considered. The energy recovery from pyrolysis was made in a horizontal tube furnace under anoxic and isothermal conditions of selected temperatures of 300 °C, 400 °C, and 500 °C. Critical metals that include the rare earth elements and other metals (such as In, Co, Li) and valuable metals (Au, Ag, Pt group) were recovered from electronic components. Pyrolysis produced liquid and gas mixtures of organic compounds that can be used as fuels. Still, the process also emitted particulate matter and semi-volatile organic products, and the remaining ash contained leachable pollutants. Furthermore, toxicity characteristics leaching procedure (TCLP) of e-waste and partly oxidized products were conducted to measure the levels of pollutants leached before and after pyrolysis at selected temperatures. TCLP result revealed the presence of heavy metals like As, Cr, Cd, and Pd. Lead was found at 160 mg/L in PCBs leachate, which exceeded the toxicity characteristics (TC) limit of 5 mg/L. Liquid sample analysis from TCLP also showed the presence of C₁₀–C₁₉ components, including benzene. This study's results contribute to the development of practical recycling alternative approaches that could help reduce health risks and

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CRedit authorship contribution statement

Endalkachew Sahle-Demessie: Principal investigator, research leader and supervision, writing. **Bineyam Mezgebe:** Post-doctoral associate conducted pyrolysis studies and manuscript preparation, TCLP and ICP-OES, writing and editing. **Joshua Dietrich:** Conducted TGA studies and FTIR work, writing, reviewing and editing. **Yonggui Shan:** Conducted on GC/MS. **Stephen Harmon:** Conducted XRF work on e-waste materials. **Chun C. Lee:** Energy calculations and valorization analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jece.2020.104943.

environmental problems and recover materials from e-waste. These results will also help assess the hazard risks that workers are exposed to semi-formal recycling centers.

Keywords

E-waste recycling; Pyrolysis; Metals recovery; VOC/PAH emission and health risks; Heavy metals; Organic pollutants

1. Introduction

The growth in information and communication technologies (ICTs) has brought a fundamental transformation in global economic, social, and political developments over the past decades. As the average useful life of electronic products decline, obsolete products are being stored or discarded with increasing frequency, creating large volumes of used electronics. The fate of used electronic devices following primary users may be reuse, storage, recycling, or disposal. Though some of these devices may find a profitable market for a second life cycle, either as a refurbished or remanufactured product, most electronic devices are recycled for material recovery, discarded, or exported [1,2]. There is an increasing need to develop technologies for the recycling of end-of-life electronic devices or e-waste, which is growing approximately 3–5% per year [3]. Thus, as the reuse and commodity market changes, sustainable management of e-waste has become a serious global challenge.

E-waste is a complex mix of high-value materials, plastics, and hazardous components that make it difficult and expensive to treat in an environmentally sound manner. Because of this, only one-fifth of e-waste is formally recycled, while the rest ends up in landfills or being informally recycled in developing countries in conditions that are hazardous to workers and damaging to the environment [4–6]. A US EPA study suggested that although it represents a small fraction of solid waste going to landfill (less than 2% of solid waste streams), e-waste may represent as high as 70% of the hazardous waste highlighting the significance of landfilling e-waste [7,8]. Proper e-waste management is thus necessary for both developed and developing countries in order to preserve the global environment and public health [9].

Besides solving the environmental and health concerns of informal recycling and landfill disposal (which can lead to leaching of hazardous elements such as brominated flame retardants and heavy metals), the recycling of e-waste has an enormous potential alternative source of valuable metals and critical materials [10–12]. Electronic devices possess a diverse composition of base and valuable metals, such as Cu, Ni, Pd, Au, and Ag, in concentrations that exceed those found in mineral ores [13]. For example, there is 100 times more gold in a ton of e-waste than in a ton of gold ore [14]. This condition has activated a significant growth of the e-waste recycling industry in the U.S., processing over 4.4 Mt of e-waste in 2011 to produce commodity products such as steel and copper [15]. Additionally, electronic devices contain small but significant quantities of rare earth elements (REE) in speakers, vibrators, and hard disks, which can also be recovered and recycled [12]. Therefore, the recycling and recovery of REE from e-waste can contribute to the stabilization of the REE market and reduce the dependence of REE mining [12,16,17]. A United Nations study stated

that global production of e-waste had reached 50 million tones, with a value of over \$62.5 billion [18].

Informal e-waste recycling, recyclers typically first shred e-waste to separate the recyclable metals with higher market reuse value from plastics. In the U.S., electronic-plastics are estimated to be 3.4M tons, comprising 5% of municipal waste [2,19]. Recycling electronics plastics has become more challenging due to minimal domestic demand, the difficulty of effective sorting, and the contamination and environmental concerns regarding flame retardants. Processors currently rely on the export market for plastics recycling, but this faces an uncertain future because of the mixed content of e-plastic and the presence of brominated (BFR), organo-phosphorus flame retardants (OPFRs), and heavy metals that have associated health concerns [20,21]. This leads the e-scrap industry to consider different end-of-life options, including converting plastics to energy and pyrometallurgical recovery of materials [22]. Thermal processing of e-waste can be developed as a safe alternative for recycling and recovery of materials while minimizing human and environmental exposures from released pollutants. Numerous studies examined pollution associated with e-waste combustion or pyrolysis and environmental impacts [6,17]. The release of hazardous substances into the atmosphere from informal recycling operations and subsequent deposition of these contaminants on soil, sediments, and water account for the indirect environmental impact [23].

This investigation's objective was to conduct experimental pyrolysis studies to recover critical materials and energy from mixed e-waste. Emission studies from pyrolysis experiments were intended to provide risks of informal e-waste treatment practices by measuring releases of both inorganic and organic pollutants. Pyrolysis and thermal analysis integrated with chromatographic and spectroscopic techniques were used to determine emissions of volatile organic chemicals and particulate matters during treatment processes. This study provides some exposure assessment that seeks to determine the intensity, frequency, and duration of actual or potential exposures to a pollutant from the e-waste recycling in the environment (Fig. 1). The data can be used to provide insight into the environmental impacts of unsustainable informal recycling practices.

Although many published studies show emissions and pollutant release from e-waste combustion, there are limited studies connecting emissions with health risks [24–29]. Facility-based risk characterization for workers and surrounding communities is a high priority issue for stakeholders. This article presents a comprehensive study on the generation of pollutants (organic, inorganic, and particulate matter) during the pyrolysis of e-waste, measured using different methods. Although many published studies show emissions and pollutant release from pyrolysis and combustion of e-waste, there are limited studies connecting emissions with exposure and health risks analysis. This study links the concentration data for each medium and combined it with information regarding exposure analysis and risk estimation.

Determining the type and amounts of emissions may also be used to assess the risk of exposure of workers at semi-formal recycling centers. Understanding the full impact of unsustainable waste management practices helps determine the human health and

environmental impact of these practices. Many recycling facilities that use manual dismantling, open burning, and acid leaching of e-waste could lead to environmental pollution and exposing workers. The toxicity characteristics leaching procedure (TCLP) data were collected on e-waste before and after pyrolysis to assess informal recycling practices' ecological impacts. Gasification and pyrolysis oil composition of electronic components of e-waste were tested for the presence of polycyclic aromatic hydrocarbons, halogenated flame retardants, and other thermal degradation products. The goal is to provide data and analysis that increases public awareness for sustainable management of used electronics and provides examples of the negative consequences of informal recycling methods.

2. Materials and methods

2.1. Materials

The composition of e-waste varies significantly, even for the same types and categories of devices that come from limited origin waste streams. E-waste consists of many components made up of more than 1000 different substances and up to 60 elements from the periodic table [6,30]. Electronics waste feedstock used in this study mainly included used cell phones and laptop computer parts. The first step for the sample collection process was dismantling used electronics products into major components. Batteries, magnets, and cathode-ray tubes were not considered for this study. Samples of electronics plastic and printed circuit board were sorted, collected, shredded by using a drill, and collecting drill bits into particles in sizes of about 1 cm. Samples were then ground and homogenized using a Micro-Mill grinder (CP Lab Safety) or a ball mill (Bel-ART products Micromill, Pequannock, NJ) to particle sizes about 1 mm. The e-waste was categorized into three classes: electronics plastic (plastic covers, housing, and structural components), plastic cable cord covers and peripheral, and printed circuit boards (PCBs). Different types of electronic plastic waste used were a Sony laptop casing, cell phone casing, HP PC desktop cover, keyboard, and plastic fan (Table 1). The plastic cable cord contains two different types of PVC cable shield (CSX-1 to CSX-5). Parts of printed circuit boards were obtained from Dell and HP computers, Intel microprocessors, and HP PC RAM.

Twenty-five samples were collected from different popular brands of used laptops and desk computers, cellular phone parts, and other peripheral items. Detailed experimental steps and outline of this project's general objectives are presented in Table 1 and in Figs. S1 and S2 of the Supporting Information (SI). Each sample was tested, and the results were categorized into three groups of the casing, printed circuit board, and electronic peripherals.

2.2. Experimental methods

2.2.1. Thermogravimetry (TG) analysis of e-waste—Thermogravimetric (TG) and differential thermogravimetric (DTG) analysis were performed to characterize the weight loss of e-waste samples as a function of temperature or time under a controlled non-oxidative environment. These tests were similar to the previous decompositions studies of solid and liquid waste materials using the TGA technique [31–34], e-waste [1], and printed circuit boards [35].

TGA measurements were performed on a Pyris 1 TGA thermal analyzer (Perkin Elmer, Sheldon, CT, USA). Different parts of electronic components were shredded into 1–2 mm size pieces and homogenized using a micro mill before placed in a platinum pan for TGA measurement. A small amount (2.1–6.2 mg) of the samples were placed in a platinum crucible pan hanging on a sensitive microbalance. Three replicates tests were run from each sample group. The sample was then heated in the furnace from an initial 30 °C to a maximum 800 °C temperature at a heating ramp rate of 10 °C /min under a helium flow rate of 20 ml/min to remove all gases evolved and avoid thermal degradation.

A continuous gravimetric record was taken as a function of time and temperature while the thermograms were treated using the Pyris Manager Software, accompanying the instrument [36]. For polymeric materials, the weight loss characteristics were compared with pure resins to compare decomposition, oxidation, or loss of volatiles characteristics [33]. Tests at different heating rates were also shown a change in mass determines the chemical composition, thermal stability, number and sequence of reactions and kinetic parameters [34]. The purpose of this study was to experimentally investigate the thermal decomposition and pyrolytic characteristics of electronic components. The compositions of traces residue of the samples left during thermal treatment of e-waste components were analyzed according to the method from Gil et al. [32].

2.2.2. Thermogravimetric analysis in tandem with GC-MS—Thermal degradation comprises a complex set of reactions involving the formation of radicals and breaking the polymer bonds. The reactions generate free radicals as primary products, resulting in an array of secondary reactions, liquid and gaseous products, and leaving small quantities of char or ash. The combination of TGA/pyrolysis and GC-MS was a useful tool for determining polymers' degradation pathways [33]. The TGA was coupled to the GC-MS spectrometer via a heated fused silica capillary transfer line with a temperature and gas flow control module (PerkinElmer TG-GC-MS system Fig. S3). This heated line was used to transport the volatile products that evolve during the decomposition of the sample at the TGA pan. The collection mechanism enables continuous detection of the MS spectra of emitted gases [34].

2.2.3. Pyrolysis of e-waste components in a tube furnace—Pyrolysis of e-waste consists of polymeric resins' thermal degradation under anoxic conditions or in the absence of oxygen or air at 350–800 °C [37]. In the pyrolysis process, the polymeric materials are heated to high temperatures, so their macro-molecular structures are broken down into smaller molecules, and a wide range of hydrocarbons are formed. Depending on the type of e-wastes, these pyrolytic products can be divided into a gas fraction, liquid fraction, and solid residues. The experimental laboratory system for pyrolysis and thermal oxidation of e-waste is shown in Fig. 1.

Approximately 1–2 g of samples were ground and homogenized before the pyrolysis experiments were conducted in a horizontal tube furnace at three temperatures of 300, 400 and 500 °C (all ± 5 °C). Fused quartz furnace tubes were used (Carbolite Tube Furnace TZF 12/38/400) with a maximum temperature of 1200 °C, with 400 mm heated length, Tube 450 $\times$ 38 mm, and three heating zones. Gasification and pyrolysis reaction were performed by

placing samples into a ceramic crucible (19 mm × 100 mm × 10 mm) in stainless steel reactor, with an internal diameter of 22.5 mm, as shown in Fig. 2. The reactor was placed in a tube furnace, where temperatures for the different zones were measured with 12-channel K-type thermocouples (RDXL12 SD, Omega, Stamford, CT, USA) and data were registered with an SD card. Samples of vapor emissions were collected using 75 ml Swagelok cylinders that are made of 316 L stainless steel (Restec-24130-PI gas). The samplers included a quick connect on the one end, and a valve with rubber septum at the other end where gas samples were drawn with a syringe for GC-MS analysis.

2.2.4. Fourier transfer infrared spectroscopic (FTIR) and x-ray fluorescence (XRF) analysis—

Attenuated total reflectance (ATR) in combination with infrared spectrometry that uses a Fourier transform FTIR spectrometer (Agilent Cary 660) with Cary 620 FTIR microscope was used to identify polymeric components and measure effects of thermal oxidation of e-waste. The system uses a helium-neon laser operating in the visible region at 632.8 nanometers. The ATR – FTIR was used to collect spectra from 4000 to 650 cm^{-1} with a data interval of 1 cm^{-1} . The resolution was set at 4 cm^{-1} . The ATR diamond crystal was cleaned with 70% 2-propanol, and a background scan was performed between each sample. Each sample was compressed against the diamond with a force of at least 80 N to ensure good contact between the samples and the ATR crystal. The absorption bands were identified using a peak height algorithm within the Bio-Rad spectra library. A four to six matching absorption bands were required for acceptable identifications.

In addition to the FTIR spectroscopic measurements, the crushed and milled printed circuit board (PCB) from different electron devices were analyzed using x-ray fluorescence (XRF). XRF was used to determine their chemical composition from beryllium (Be) to uranium (U) in concentration ranges from 50+ wt% to sub-ppm levels. The tests were conducted on pyrolyzed and TCLP analyzed samples. E-waste samples were ground in a tungsten carbide ball mill before the powder samples were sieved through a 200 mesh (<75 mm) screen. Not all components of the e-waste were milled equally since some malleable components such as copper circuit-wiring tended to flatten. The sieved powder was mixed with a binder, pressed under 30 tons pressure to form a pellet, and then analyzed by PANalytical Axios Advanced wavelength dispersive x-ray fluorescence spectrometer.

2.2.5. Analysis of pyrolysis products: gas phase and liquid phased samples

—Gas emission from pyrolyzed e-waste samples at different isothermal conditions was collected using stainless-steel cylinders, from which samples were drawn gas-tight syringes and manually transferred to the injection port of a gas chromatograph (Perkin Elmer Clarus 600). The GC was equipped with a capillary column (Agilent DB5, 30 m × 320 μm , i. d. = 0.25 μm film thickness). The injection port was initially held at 35 °C for 2 min, then raised at 35 °C/min up to 230 °C and held constant. The helium carrier gas pressure was held 5 psi. The oven temperature was initially held at 35 °C for 1 min, then raised at 2 °C/min up to 60 °C and then at 25 °C/min up to 120 °C and held for one min. Samples were detected using a mass spectrometer (PerkinElmer Clarus 600T). The MS analysis settings were set at acquisition Scan mode with a solvent delay of 0.1 min and mass scans ranging from 50 to 500 mu. The gas samples were analyzed at 300, 400, and 500 °C for each e-waste sample.

Liquid samples were collected from the sample pan at the end of each pyrolysis test and rinsed with methanol and placed in 20 ml vials. The extracted liquid samples were then analyzed using a GC-MS (Agilent Technologies 6890 GS-MS of 5973). The oven temperature was initially held at 45 °C for 1 min, then ramped up at 25 °C/min up to 120 °C and then raised 290 °C at 8 °C/min and held for 3 min. The sample was then eluted through a column (DB5, 30.0 m × 250 µm i.d 0.25 µm film thickness) with splitless inlet mode. The column flow rate was set at 1 ml/min, with an average velocity of 26 ml/s.

2.2.6. Measuring particulate matter emissions—At selected pyrolysis temperatures, particulate matters emitted were measured using a laser light scattering airborne particle counter (LSAPC, Aerotrak Handheld Particle Counter 9301–01, TSI, USA). The particle count was converted into a mass over volume (µg/m³), applying a bulk density of the particulate matter 1 g/cm³ over a nominal volume of each dimensional class [38]. The particles were counted in three channels between; 0.3, 2, to 5 µm. The model has a 2.83 L/min flow rate (2,83 LPM) [39]. After a warm-up period of 15 min, particle count was taken at room temperature, 300, 400, and 500 °C right before gas analysis. The detector complies with the stringent requirements set by ISO-21501–4 and calibrated with the National Institute of Standards and Technology traceable polystyrene latex spheres.

2.2.7. Leaching tests used electronics and pyrolyzed products—The toxicity characteristic leaching procedure (TCLP) was conducted to evaluate the different contents of organic and inorganic compounds that may leach if e-waste or pyrolyzed materials are left in the environment. The TCLP is designed to simulate landfill leaching under a worst-case scenario. The leaching tests are reported in mg per liter and are usually used as a “screening” technique to demonstrate that waste is considerably above or below a threshold. Earlier TCLP studies on e- waste used samples of the whole product rather than its parts, which may not adequately reflect each component’s individual toxicity [40].

In this study, the EPA Method 1311- standard TCLP method was employed to the e-waste to determine the mobility of organic and inorganic analytes [41]. The method includes crushing samples to particles less than 6.3 mm, adding an extraction fluid at a 20-to-1 fluid to-sample ratio, and rotating the sample in extraction fluid for 18 h. One liter of TCLP extraction solvent consists of 5.7 ml glacial acetic acid, 64.3 ml 1 N NaOH, and 930 ml of Milli-Q water, while the pH of the extraction fluid was set 4.93 ± 0.05. The tests were intended to show whether disposal management choices, such as pyrolysis, affect the total waste leachate. These choices can be compared to appropriate regulatory limits for specific organics and elements As, Cd, Pb, Hg, Se, and Ag. After sample filtration (0.45-µm membrane filter) heavy metals in the leachate were determined by inductively coupled plasma-atomic emission spectroscopy (ICP-OES, PerkinElmer; U.S. EPA Method 6010B), after metals digestion for liquid samples (U.S. EPA Method 3020 A). Volatile organic compounds were analyzed by gas chromatography coupled with mass spectroscopy (Agilent Technologies 6890 GS-MS of 5973) [42].

2.2.8. Inductively coupled plasma - optical emission spectrometry—The ICP-OES (Optima™ PerkinElmer) was used for the elemental analysis of the TCLP extracts. The instrument is a simultaneous ICP with an echelle polychromator and a Segmented Array

Charge-coupled Device (SCD) detector. Simultaneous measurement of background analyte emission allows for accurate correction of transient background fluctuations. The instrument parameters were optimized to detect As, Cd, Pb, Hg, Se, and Ag. In most cases, As, Cd, Pb, Hg, Se, and Ag were present in the matrix; therefore, a blank subtraction was used. The instrument conditions used for the determination of the method detection limits and analytical results are shown in Table S3. Calibration standards were prepared from PerkinElmer Pure TCLP Multielement Standard (PerkinElmer Part No. N930–0241).

Three standards and a blank were used for calibration. Calibration standards were made in HNO₃/HCl matrix, as specified in Method 6010, to approximate the digested form of the samples. Method 6010A, Revision 1 [42] was followed for the analytical determination of the TCLP elements required in the extract. Method 6010A is a Resource Conservation and Recovery Act (RCRA) program method published in Test Methods for Evaluating Solid Waste, Physical/Chemical Methods, publication SW-846, with revisions (2). The instrument was calibrated with a filtered matrix blank and dissolved As, Cd, Pb, Hg, Se, and Ag matrix spikes (0–20 µg/L spikes). The system was equipped with a concentric nebulizer, cyclonic spray chamber, and platinum sampler and skimmer cones. The reference power was set to 1600 W to ensure the complete atomization of nanoparticles. Parameters optimized daily included the sample flow rate (between 0.26 and 0.29 ml/min) and the transport efficiency (between 7.5% and 8.5%). Masses of the given metals were monitored with a dwell time of 100 µs over a 100 s sampling time. The extracts of different parts of used electronics and pyrolyzed e-waste will be digested 10% HNO₃/HCl in Teflon bottles. The first wavelength given for each element is recommended in EPA Method 6010. The second wavelength is an alternate emission line that was monitored to identify potential interferences.

3. Results and discussion

3.1. Thermogravimetric (TGA) and differential thermal analysis (DTA) results

TGA and DTA analysis of electronic peripheral, electronic casing plastics, and printed circuit boards (PCBs) were conducted under a helium atmosphere. The tests for the furnace temperature ranging from 30 to 700 °C at a heating rate of 10 °C/min are shown in Fig. 3(a)–(f). The TG analysis of the various e-waste samples was used to measure the onset and the maximum weight change, which is the DT's curve inflection point curve and the char weight percent at 600 °C. All samples showed a small, low-temperature weight loss step up to a minimum of 330 °C due to the release of water and volatile components. The main decomposition steps started at about 363 °C and ended at about 481 °C. The nature of thermal degradation differed significantly depending on the electronic component and polymeric composition. Thermal degradation of polymeric casing materials is usually fast and occurs from 320 to 340 °C. The extrapolated onset temperature is when the weight loss begins, which is a reproducible temperature calculated according to ASTM E1131–08 [43]. These results are summarized in Table 1. The onset of the weight loss for the TG curves also reflects the stability of the samples. Samples from polymer cases had initial decomposition temperatures 317 and 298 °C (Fig. 3a and b), whereas printer circuit samples start to decompose at higher temperatures of 362 and 413 °C, respectively. The range of onset temperature varied from 224 to 422 °C for cable cord covers and PCB, respectively. The

minimum onset temperature was observed for a Sony laptop casing, and a higher onset temperature was for a cell-phone casing.

The maximum weight loss, indicative of major degradation of e-waste samples, occurred at 413 ± 84 °C. The maximum weight loss temperatures for computer casing, peripheral, and PCB samples were 536, 441, and 526 °C, respectively.

For most of the electronic plastics materials, the average temperature for a 50 wt% loss was 474 °C, close to 480 °C, when the maximum decomposition rate occurred. The highest e-waste plastics degradation temperature recorded was for composite laptop casing at 537 °C. The minimum percent of ash remaining for this category of e-waste analysis was 1.5% for the HP PC keyboard. The decomposition step for cable cover polymer (PVC) started around 223 °C and ended at about 310 °C. The first onset temperature was around 223 °C, and the second onset was at around 450 °C, with a maximum degradation rate at 466 °C. The average 50% and maximum decomposition temperatures for the cable cords were similar at 470 °C. The ash remains for polymeric components vary from near zero to as high as 28 wt % for plastic air fan and 33 wt% for both black and gray cable cover. Literature data show that most of the degradation occurs between 450 and 550 °C [1,44]. Fig. 3(b) presents the TGA analysis of the cable cords coatings that are mainly made of PVC. The thermal degradation process of these cable cords (mainly PVC materials) is a two-step process, which has been discussed by different studies [45–47]. The dehydrochlorination of the polymer and the formation of conjugated double bonds happen in the first step. The second step includes the formation of low concentrations of hydrocarbons with linear or cyclic structures and the polymer's continuous degradation [48].

The TG plots for polymer casing samples were smooth with single DTG curves and close to 100% weight loss at 600 °C. Pyrolysis of PCB samples shows a multistage break down that may include devolatilization and char pyrolysis at 444 °C (Fig. 3c and d). After a rapid loss of weight and following the thermal degradation emission release, 50+ wt % residual mass remained. The thermal degradation profile for the various components of e-waste was different. The DTG curve shows the heterogeneity of printed circuit board samples. All the samples left residues depending on their source, consisting of plastics or printed circuit board [49]. The decomposition of PCBs occurred between 323 and 366 °C [50]. The decomposition temperature of e-waste was much lower than most polymers typically found in municipal waste, which is in the range of 400–500 °C [50,51].

3.2. Weight loss during pyrolysis of e-waste

The weight loss measurements for larger volume e-waste samples during pyrolysis in the tube furnace at selected temperatures from 300 to 500 °C confirmed the TGA data on the progressive weight loss. Although volatile organic compounds start to be released at temperatures below 300 °C, the main weight loss was registered between 300 and 400 °C, which was attributed to the thermal degradation and depolymerization of plastics components and the loss of semi-volatile compounds. The weight decrease observed from 400 to 500 °C was assigned to the volatilization and pyrolysis of heavy organic compounds denominated fixed carbon; the residue non-combusted above 500 °C that presented a stable weight was ashes.

Fig. 4a shows the weight loss of casing materials, including casings of laptops, cellphones, and desktop computers. Pyrolysis of laptop computer showed a 60% weight loss at 500 °C, whereas the desktop cover showed a 95% thermal degradation at 500 °C with no char formation. A similar change in the chemical bonds in their molecular structures and degradation mechanisms based on TGA results was observed by other researchers [52].

The weight loss for plastic peripheral (Fig. 4b) shows that the thermal degradation for both gray and black cable covers show a similar reduction of 20% at 300 °C. At 400 °C pyrolysis, the weight of PVC-CSX cable cover samples were reduced by 40% and 60%, and at 500 °C were 60% and 80%, respectively [53]. The maximum degradation was observed for poly (methyl methacrylate) cooling fan at 95% pyrolyzed at for 500 °C temperature, which left no residues (Fig. 4b). Samples leave residues depending on their source, consisting of plastic-based or PCB samples [49,50]. The highest weight loss for PCBs occurs between 323 and 366 °C due to the pyrolysis of the light fraction of the printed circuits (Fig. 4c) [50]. The peak degradation rate of electronic polymers in this study was measured at temperatures lower than most polymers typically found in municipal waste, ranging from 400 to 500 °C [50,51].

3.3. Material characterization with FTIR and x-ray fluorescent spectroscopy (XRF)

Samples of e-waste materials before and after pyrolysis at selected temperatures were analyzed using FTIR and XRF. FTIR spectra library (Bio-Rad) was used to identify the main polymeric materials (Table 2). Furthermore, images showing physical changes were taken using a digital optical microscope and scanning electron micrographs samples due to pyrolysis at selected temperatures are given shown in Figs. S5 and S6 in supporting materials, respectively.

The IR spectra of e-waste materials before and after pyrolysis are shown in Fig. 5(a)–(d). Adsorption peaks at 829 and 1505 cm^{-1} shown in Fig. 5(a) are characteristic of 1, 4 substituted aromatic rings- stretching from polycarbonate or polystyrene type thermoplastics. The absorption bands around 1188 and 1080 cm^{-1} are characteristics of the para aryloxy group's presence on polycarbonate, while the peaks between 1615 and 1495 cm^{-1} suggesting phenoxy aromatics. For compounds with C–H stretching, show peaks near 3000 cm^{-1} .

Fig. 5 shows the FTIR spectra of the different components of e-waste during thermal treatments. Fig. 5(a) represents the ATR-FTIR spectra of cellphone casing at different temperatures (at room temperature, 300, 400, and 500 °C). The absorption bands identified for ABS material were closely matched by the absorption bands shown in Fig. 5(a). As shown in FTIR spectra of the different components of e-waste during thermal treatments (Fig. 5). The presence of acrylonitrile butadiene styrene (ABS) was ascertained by the spectrum of C H stretching at 3025 cm^{-1} (aromatic), at 2850–2923 cm^{-1} (CH_3 and CH_2), and C H bending in mono-substituted benzene rings at 759 and 700 cm^{-1} . The band at 1600–1602 cm^{-1} was assigned to C-C stretching aromatic from polystyrene. The characteristic polyacrylonitrile (C–N bond) is determined based on the spectrum at 2238 cm^{-1} and the butadiene region occurs at near 966 cm^{-1} . From the ABS spectrum, the bands at 1732 and 1157 cm^{-1} show the presence of the CO (the acrylate carboxyl group) and C–O–C

stretching vibrations, respectively. Selected absorption peaks with their functional groups were located at approximately 2968 (aliphatic methylene), 1770 ($\text{C}=\text{O}$ stretch), 1504 (aromatic ring), 1188 (polysaccharides) and 829 (carbonate) [54]. IR spectra of the electronics component before pyrolysis has a close similarity to polycarbonate and ABS [54]. The high temperatures of pyrolysis (400 and 500 °C) resulted in the loss of most functional groups.

On the other hand, Fig. 5(b) shows the spectra of a keyboard at different temperatures. The spectra at room temperature compared to the spectra of originally manufactured material (mainly polycarbonate and ABS), the appearance and number of identifiable wavenumbers were nearly identical [54]. As shown in the figure the peaks with their functional groups were located at 2922 (aliphatic methylene), 1493 (aromatic ring), 1452 (CH_2 bend), 752 (CH out-of-plane bend, $-\text{CH}$ bend) and at 696 (aromatic CH out-of-plane bend) [54,55]. Similar to casings, pyrolysis at temperature of 400 and 500 °C degraded all the polymers' functional groups.

Fig. 5(c) presents plastic cord cover's (mainly PVC material) IR spectra with main peaks and functional groups at room temperature. FTIR spectrum for the pyrolysis of cable coating material, mainly PCV shows the spectrum band of $2400\text{--}2260\text{ cm}^{-1}$ is associated with CO_2 . Formation of hydrogen chloride (HCl) can be observed in the band range of $3100\text{--}2600\text{ cm}^{-1}$, and the absorption band of H_2O corresponds to $1800\text{--}1300\text{ cm}^{-1}$ and $4000\text{--}3500\text{ cm}^{-1}$ (Fig. 5(c)). The spectrum band in the range of $700\text{--}550\text{ cm}^{-1}$ is attributed to C-Cl stretching vibrations. The results show that the gases from the aged cable sheath are released later, but more quickly, than those by the new one, especially HCl gas. At increased temperatures, the C-H, $\text{C}=\text{O}$ and C-O groups disappeared after the pyrolysis. The FTIR spectrum of PCBs at different temperatures is shown in Fig. 5(d), confirming heterogeneity of PCBs' materials with complex structures that changed gradually in the pyrolysis process [56, 57]. The prominent peaks for PCB at room temperature were at 3386 (O-H, hydroxyl group), at 2927 (C = O, methyl and methylene group), at 1072 (C-O, alcohol R-OH group) and 800 (C-H, benzene derivative group) [58]. After the pyrolysis, all functional groups except Si-O group had not been detected.

Wavelength-dispersive X-ray fluorescence (WD-XRF) spectrometry was used to determine the elemental composition of ground PCB samples before and after pyrolysis at selected temperatures. A total of 35 elements were detected that include heavy metals (As, Cr, Cd, Pb), rare earth elements (57–71 on the Periodic Table, e.g., La, Nb, Y), platinum group elements (Pd), other metals (Cu, Fe, Zn, Zr, Co) and non-metals (Br, Cl, I, S, O, P) (Fig. 6(a)). As the PCB was pyrolyzed, the percent fraction of recoverable materials increased significantly with the pyrolysis temperatures (Fig. 6b).

3.4. Gas and liquid pyrolysis products

Pyrolysis of e-waste was carried out under a nitrogen atmosphere in a tube furnace, and gas and liquid fractions were analyzed with GC-MS. The GC-MS analysis of the liquid fraction, pyrolysis oil, contains high concentrations of phenolic oils and hydrocarbons (Table 3). The composition includes $\text{C}_8\text{--C}_{10}$ compounds such as styrene resulting from depolymerization of polystyrene, brominated phenols and a mix of benzoic and hydroxybenzoic acids. The

average molecular weight of the e-waste pyrolysis oil obtained from electronics casings, PCB, and peripheral were about 136 ± 47 , 173 ± 48 , and 183 ± 55 g, respectively. The results agree with literature reports for polycarbonate/polyamide with similar properties to motor gasoline-like fractions [1].

Analysis of gas-phase pyrolysis products released during the pyrolysis of different components of e-waste was carried out using GC/MS (Perkin Elmer Clarus 600). The thermal decomposition of PCB substrate scrap consisted mainly of C₆ benzene, bromo (C₆H₅Br), C₆ (phenols and derivative), C₇ (toluene, benzaldehyde and p-cresol), and C₆–C₁₀ components such as styrene and benzofuran, 2-methyl (C₈H₈O) (Table S5). The primary components were benzene, styrene, phenol, benzaldehyde, phosphonic acid, phenol, 2-bromo and C₉ benzofuran, 2 -methyl. The pyrolysis products of PCB waste were composed of a mixture of aromatic derivatives, aromatic amines organic compounds of C₆–C₁₆ and brominated organics. The detection of compounds such as aromatic and phenolic compounds gives evidence of bisphenol-A occurrence used in the epoxy formulation [38,39,41,45,46,59]. Pyrolysis products of unbrominated epoxy resin used for PCBs are substituted phenols and aromatics, indicating a favored splitting of the O–CH₂ and C(phenyl)–C bonds [47–49].

The identification of pyrolysis products released from the thermal degradation of plastic samples from cellphone and laptop cases was carried out using pyrolysis-GC/MS. The gas and liquid analysis results are presented in Table 3 and 4. The products derived from plastic components of e-waste consisted of benzene, toluene, styrene, phenol, α -methylstyrene, acetophenone and phenol, 2 – bromo (Table S5). Additional examples of the chemical mixtures of gas emissions for PCB of cellphones is presented in Fig. S4 of the Supporting Information.

The relative abundance of emitted pyrolysis products was compared. Comparing the pyrolysis characteristics of PCB substrate and plastic components of e-waste scrap revealed that 2-methyl, methyl ester 2-propionic acid was the main product rather than aromatic compounds. Moreover, the relative amounts of some products (e.g., C₄ [1,3-butadiene and 2-methoxy-1-propene]) were significantly higher than emitted PAHs. The cellphone casing was identified as polymethyl methacrylate. Thermal degradation of polymethyl methacrylate around 340 °C is the chain radical, forming methyl methacrylate as a volatile product. The initiation of PMMA depolymerization proceeds via disruption of the link involved in the formation of free radicals at a first-order rate. At temperatures between 340 and 360 °C the thermal degradation of PMMA was initiated by a mixture of chain end and chain scission processes, followed by depropagation [58].

The pyrolysis study was carried out at temperatures 300, 400, and 500 °C to study the evolution of decomposition byproducts of e-waste compounds. Following the TGA data, at 550 °C the decomposition of all the compounds was nearly completed. Hence, the results obtained mainly to show the primary products and direct derivatives that are formed in the decomposition process. Particle count has been carried out at each specified temperature before gas/ liquid analysis. During gas analysis, the emitted pyrolysis products released from the pyrolysis of e- waste are presented in Table 3. The distribution of the molecular weight

of emitted VOCs at selected temperatures is presented in Fig. S7 of Supporting Material. The average molecular weight of the e-waste pyrolysis oil obtained from electronics casings, peripheral and PCBs were about 122 ± 37 , 134 ± 33 , and 108 ± 36 g, respectively. The pyrolysis products derived from the e-waste consisted mainly of benzene, phenol, toluene, C₆–C₁₀ components, acids, aldehydes, and brominated compounds. The primary components were benzene, styrene, phenol, toluene, and 1,3-dimethyl-, ethylbenzene, p-cresol, and different xylenes.

Table 4 presents the emitted byproducts obtained from the liquid analysis. The identification of pyrolysis products released from the thermal degradation of plastic casings, peripheral, and PCBs were carried out using GC/MS (Fig. S6). The products derived from the e-waste samples are identified and listed in Table 4. The relative abundance of emitted pyrolysis products was compared, which showed the pyrolysis products mainly consisted of styrene, phenol, p-cresol, C–C₁₈ components, brominated compounds, phenol, bisphenol - A, and benzoic acids.

3.5. Particle emission from pyrolysis

Particle measurements were taken using a particle counter Aerotrak (Handheld Particle Counter 9306, TSI, USA) at different temperatures to investigate the emission of particulate matter during pyrolysis of e- waste. Fig. 7 presents particle emission during pyrolysis for different category of the e-waste. In the Fig. 7 the X- axis indicates the temperature variation, left vertical axis indicates the particle count per cubic meter, and the right vertical axis indicates the emission by mass (μg) per cubic meter. Fig. 7a shows most of the emissions during the pyrolysis of cell phone casing occurred at about 300 °C. As the temperature increased, the number of higher particles started emitting. Particulate emission rate agreed with TGA observation on weight loss that showed most of the weight loss was between 300 and 400 °C. The amounts of larger particles (5 μm) also increased from zero in the count at room temperature to 25,559/ m^3 at the end of 500 °C pyrolysis temperature. Fig. 7b presents the particle count result for the pyrolysis of desktop cover. As a result, all particle sizes started emitting at 300 °C. The emission of 5 μm size particles was highest at 300 and 400 °C. Size 3- μm emitted concentrations were 440,047 / m^3 at a higher temperature of 500 °C, Fig. 7c shows the particle count for plastic cable cover material. A higher count was observed for smaller size particles at 500 °C (1,255,242/ m^3). It showed that cable materials produce smaller particulate matter at this given temperature. More 5 μm sized particles were released at 300 °C. The particle count results for PCB are presented in Fig. 7d. In addition, the pyrolysis of electrical cord covers, which are mostly PVC, resulted in higher levels of 3 and 5 μm particles (Fig. 7d).

Pyrolysis of PCB resulted in the emissions of particulate matters in 3 and 5 μm range at 400 and 500 °C; the number of particles in the 0.3 μm range was negligible (Fig. 7c). The TGA peaks for cable coating plastic cord correspond to the typical peaks found on PVC polymer [47,54,55, 60]. Following the pyrolysis of the cable cover material, all surface functional groups in the FTIR spectra disappeared except the carbonates at 1257 cm^{-1} , which were present until the end (Fig. 5(a) and (b)). The major pyrolysis product of PVC in the first-stage was hydrogen chloride [61].

The pyrolysis products of the second stage are very complicated. Previous literature has stated that in this stage, a substantial quantity of hydrogen chloride and the aromatic groups were detected [60,61]. Other studies state that some weak points of C–H, C–C, and C–Cl in PVC chains indicate the degradation of PVC polymer [47,48]. The corresponding weight losses due to the thermal degradation of plastic cables were 20%, 40% and 60% at 300, 400 and 500 °C, related to the peaks' disappearance. This also relates with the emitted particle sizes and concentration. As the pyrolysis temperature increased, more particles also were produced (Fig. 7c). On the other hand, pyrolysis of PCBs at 300 °C removed the hydroxyl and the methylene groups. However, all but R–OH functional groups had disappeared after the pyrolysis of 400 and 500 °C. The main products of the pyrolysis oil of PCB were aromatic compounds, including substituted benzenes. The solid products mainly contained char and fiberglass [40]. Due to the metal content, the weight loss for PCB material was less than 20%.

3.6. TCLP results

The toxicity characteristic leaching procedure (TCLP) was used to characterize the potential of e-waste to leach when it ends up in the environment or disposed of in a landfill following the guidelines provided by Resource Conservation and Recovery Act (RCRA). The US EPA exempts as certain widely generated wastes as “universal waste” — such as thermostats, batteries, and fluorescent lamps — containing hazardous materials from having to meet RCRA's hazardous waste. However, e- waste components such as cathode ray tubes are not permitted in landfills, and studies indicated that e-waste fails to meet the RCRA definition of hazardous waste [62].

Electronic devices that contain circuit boards have the potential to exceed the TCLP limit for hazardous material. E-waste that breaks, corrodes, or is pyrolyzed for product recovery could release more hazardous components over time to the environment. The potential for disposal of e-waste is determined by its toxicity characteristics (TC). For ground PCB samples, tests were conducted before and after pyrolysis at selected temperatures. The results of the elemental composition of heavy metals are presented in Fig. 8. The major heavy metals detected were As, Cr, Cd, and Pd. Lead was found at 160 mg/L in PCB leachate which exceeded the toxicity characteristics (TC) limit of 5 mg/L. The inductively coupled plasma atomic emission spectroscopy (ICP-AES) results are summarized in Fig. 8 and show that the concentrations of heavy metals per weight of samples increased as the pyrolysis temperature increase from 300 to 500 °C. The full results of the TCLP analysis is shown in the Supporting Information, Table S3. The hydrocarbon compounds obtained from the TCLP analysis using GC–MS are shown in Table S4, Supporting Information.

E-waste devices are considered Universal waste requiring proper recycling since they leach enough lead under TCLP, even though ferrous metals hinder the leaching of Pb [62]. Thus, devices that contain little of no ferrous materials are more likely to exceed TC thresholds [63,64]. However, the composition of electronics has been changing significantly and levels of lead used in electronic circuits has been reduced from many devices. This study has shown that the levels of heavy metals differ considerably depending on the source, and that pyrolysis alters the TCLP results.

3.7. Energy recovery

Material and energy recovery using thermal treatment processes can be an integral part of the end-of-life management of e-waste. XRF analysis of ground samples and acid extracts of PCB samples followed by ICP-AES analysis showed that PCB contains high amounts of base metals, precious metals, and rare earth elements (Table S6). The current metal-centric recycling technologies are widely based on pyrometallurgy. Modeling calculations were made to estimate the combined heat and power generation from e-waste. The process may be used for metal recoveries accompanied by recovering monomers, such as styrene from polystyrene, or more valuable polymers such as polycarbonates [65]. Emerging energy recovery technologies are allowing innovative manufacturers to derive fuel from plastic waste and develop new feedstock materials for manufacturing [1,65,66]. The recovery of polymer material or energy from e-waste includes alcoholysis and hydrolysis [65, 66]. Chandrasekaran et al. recovered up to 87% polycarbonate from cell phones and other e-waste streams by using solvent extraction [1].

The adiabatic heat values of polymer resins commonly used in consumer electronics vary between 44M J/kg for polyolefins and 16M J/kg for PVC. Heat values of electronic plastics' mixtures range between 25 and 35M J/kg (Table S2). The energy recovery from pyrolysis and pyrolysis of e-waste plastics as a function of pyrolysis temperature and heat content to produce heat and electricity outputs for electronic components, including casing and peripheral, is shown in Fig. 8. The energy recovery is strongly related to the fuel's heat content and the operating temperature recovery furnace. This confirmed early observation by Cui and Zhang that if a comprehensive emission control system is installed, thermal processing of e-waste provides a possible recovery of energy [67]. The details about the model developed for calculating waste-to-energy are presented in the Supporting Materials' energy recovery section (Fig. 9).

3.8. Risk assessment

Air quality studies near illegal e-waste recycling and burning sites showed increased levels of particulate matter (PM₁₀) concentration [68]. Particulate matter containing heavy metals was detected near burning sites that use crude and rudimentary e-waste processing methods. Based on emission measurements from different e-waste components, estimates of risks due to chronic exposure to polluted air near e-waste processing were made [68]. Exposure to e-waste and emissions from the pyrolysis of e-waste is a complex process. The exposure routes, the duration of exposure, and proximity to the source should be considered and the possible other inhibitory effects or add to the impact. He et al. assessed that heavy metal pollution included As, Cr, Cd, Ni, Pb, Cu, and Zn at e-waste recycling activities at the Pearl River Delta [64]. They showed that the ecological risks were dominated by low levels of Cd released. Alarming levels of particulate matter and volatile organic compounds emissions were measured from the recycling process and during pyrolysis or combustion. Many of these pollutants are lipophilic, bioaccumulative substances that are persistent in the environment (Tables 3 and 4). Many of them are associated with a significant prevalence of thyroid and lung functions and other health impacts [9]. Exposure to e-waste could occur as a result of formal or informal recycling operations and hazardous compounds remain in the environment. Occupational and environmental exposure around informal recycling

operations that have persisted in some developing countries. The hazard for workers and the public who reside near a recycling location can be estimated using the equation:

$$HQ = \frac{EC}{RfC} \quad (1)$$

where HQ is the hazard quotient, EC is the exposure concentration, mg/L and RfC is the reference concentration (mg/L). RfC = No Observed Adverse Effect Level NOAEC/UF (uncertainty factor).

A hazard quotient (HQ) is the ratio of the estimates of the exposure to a substance and the levels at which no adverse effects are expected. Risk assessment due to exposure to emissions to pyrolysis that include particulate matter and VOCs was carried out to estimate chronic risk of workers exposed to the VOCs and dust that traders and consumers are exposed to at an informal electronics recycler [69] Exposure levels were estimated based on the average daily dose (ADD) of VOC and particulate matter inhaled or ingested and was determined using the equation below:

$$ADD = \frac{\text{Intake Dose}}{\text{Body weight} \times \text{Average time}} \quad (2)$$

$$\text{Intake Dose} = C \times IR \times ED \quad (3)$$

where C is the concentration of toxic pollutants (mg/L), IR = intake or inhalation rate. Chronic exposures can be calculated by summing across each life stage specific ADD. Thus, EF = exposure frequency of 260 working days per year, ED = exposure duration (hr) 6 for children and 24 years for adults, B_w = body weight of 65 kg for adults and 15 kg for children and AT = average time of 260 days $\times$ 5 yrs = 1300 [70].

We used the U.S. EPA's methodology for estimation of inhalation reference concentrations ($RfCs$), also called reference dose, RfD , as benchmarks to estimate the quantitative dose-response assessment of chronic noncancer toxicity for individual inhaled chemicals. The method helps to estimate noncancer toxicity risks that have adverse health effects other than cancer and gene mutations [42].

Table 3 shows estimated average chemical intake doses ($\mu\text{g}/\text{m}^3/\text{day}$) and target hazard quotients (THQs) caused by exposure in open burning informal recycling facilities. The toxic compounds that are found in e-waste are associated with many health risks. Fig.10 shows the HQ values of adults chronically exposed to toxic emissions of different components of e-waste. In inhalation and ingestion exposures, the dose-response parameters for carcinogenic risks should be adjusted for the difference in absorption across body barriers between humans and the experimental animals used to derive such parameters. However, this study's exposure assessment was based on the intake dose, with no explicit correction for the fraction absorbed. However, the exposure assessor needs to make such an adjustment when calculating dermal exposure. In other specific cases, current information indicates that the human absorption factor was used in the derivation. Further studies are needed on simultaneous exposure of e-waste recycling to many of the different chemicals

found in e-waste. Additional health risk assessment information on the human health and environmental implications of improper management of e-waste is given in Fig. S9 in the health risk assessment section of the Supporting Information.

4. Conclusion

In this study, the use of pyrolysis was investigated as a possible approach to recovering material and energy from e-waste in reducing electronic materials ending up in landfills. Components of e-waste contains high levels of halogenated compounds, such as brominated flame retardants, Pb, Cd that can leach out in a landfill as a complex mix of pollutants. The major findings of this study include:

1. The study confirmed that pyrolysis could be a useful technique for gasification and liquefaction of polymer fraction of e-waste and facilitating the recovery of metallic elements. Thermogravimetric studies provided more control of the reaction conditions and quantifiable information on the decomposition process of e-waste materials and toxic pollutants' emission.
2. The pyrolysis/GC/MS techniques used in this study afforded researchers to identify the composition of toxic chemicals released during informal recovery methods of precious metals from e-waste.
3. The study confirmed the environmental and human health risks of pyrolysis and open burning practiced at informal recycling operations, releasing hazardous organic and metal pollutants. The study showed that harmful compounds such as polyaromatic hydrocarbons, brominated fire retards, and elements such as Cd, Pb, and Cr are emitted at medium to high levels during e-waste processing. The exposure routes could be inhalation, dermal, or ingestion depends on the recycling process and level of technology.
4. E-waste recycling processes are essential for recovering energy (average 30M J/kg) and various precious metals. Therefore, there is potential to process e-waste to recover energy and materials. Still, strict controls such as informal recycling must be in place to reduce the risk from these harmful emissions.

However, data are minimal, and much more still needs to be learned about the extent and long-term effects of these e-waste activities on environmental and human health. The variability of TCLP results indicates the heterogeneous e-waste materials that leached significant Pb, Cu, and other elements. Emission and TCLP studies showed that various organic and inorganic pollutants, mainly Pb, could be released by informal technologies that use burning, pyrolysis, or disposal in landfills.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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References

- [1]. Chandrasekaran SR, et al., Materials and energy recovery from e-waste plastics. 6 (4) 2018, 4594–4602.
- [2]. Electronic Take Back Coalition, f.a.f. Facts and Figures on E-Waste and Recycling. 2014. http://www.electronicstakeback.com/wp-content/uploads/Facts_and_Figures_on_EWaste_and_Recycling.pdf. (Cited 25 July 2019).
- [3]. EUROSTAT. Waste Electrical and Electronic Equipment (WEEE) by Waste Management Operations. 2014. http://epp.eurostat.ec.europa.eu/portal/page/portal/statistics/search_database. (Cited 25 July 2019).
- [4]. UN. Report: Time to Seize Opportunity, Tackle Challenge of E-waste 2019. <https://www.unenvironment.org/news-and-stories/press-release/un-report-time-seize-opportunity-tackle-challenge-e-waste>. (Cited 25 July 2019).
- [5]. Diaz-Barriga F, Evidence-based intervention programs to reduce children's exposure to chemicals in e-waste sites. in: Discussion Paper for WHO Working Meeting on E-waste and Children's Health. 2013.
- [6]. Baldé C, et al., The Global E-waste Monitor–2017, United Nations University (UNU), International Telecommunication Union (ITU) & International Solid Waste Association (ISWA), Bonn/Geneva/Vienna, 2017.
- [7]. USEPA. Report: Multiple Actions Taken to Address Electronic Waste, But EPA Needs to Provide Clear National Direction. 2004. <https://www.epa.gov/office-inspector-general/report-multiple-actions-taken-address-electronic-waste-epa-needs-provide>.
- [8]. Holgate P, How do We Tackle the Fastest Growing Waste Stream on the Planet? World Economic Forum. 2018. <https://www.weforum.org/agenda/2018/02/how-do-we-tackle-the-fastest-growing-waste-stream-on-the-planet/>.
- [9]. Grant K, et al., Health Consequences of Exposure to E-waste: A Systematic Review. 1(6) 2013, e350–e361.
- [10]. Grassie N, Scott G, Polymer Degradation and Stabilisation. CUP Archive. 1988.
- [11]. Holland BJ, Hay JN, The kinetics and mechanisms of the thermal degradation of poly (methyl methacrylate) studied by thermal analysis-Fourier transform infrared spectroscopy, Polymer 42 (11) (2001) 4825–4835.
- [12]. Levchik SV, Weil ED, Thermal decomposition, combustion and flame-retardancy of epoxy resins—a review of the recent literature, Polym. Int. 53 (12) (2004) 1901–1929.
- [13]. Tanskanen P, Management and recycling of electronic waste, Acta Mater. 61 (3) (2013) 1001–1011.
- [14]. W.E. Forum, A New Circular Vision for Electronics, Time for a Global Reboot. 2019. <https://www.weforum.org/reports/a-new-circular-vision-for-electronics-time-for-a-global-reboot>. (Cited 6 August 2019).
- [15]. Friege H, Review of material recovery from used electric and electronic equipment-alternative options for resource conservation, Waste Manag. Res. 30 (9_suppl) (2012) 3–16.
- [16]. Luda M, et al., Thermal Decomposition Of Fire Retardant Brominated Epoxy Resins. 65(1) 2002, 25–40.
- [17]. Barontini F, Cozzani V, Formation of hydrogen bromide and organobrominated compounds in the thermal degradation of electronic boards, J. Anal. Appl. Pyrolysis 77 (1) (2006) 41–55.
- [18]. UNEP, UN Report: Time to Seize Opportunity, Tackle Challenge of E-waste. 2019.
- [19]. Rochman CM, et al., Policy: Classify Plastic Waste As Hazardous. 494(7436) 2013, 169.
- [20]. Yang J, Zhao Y, Li M, Du M, Li X, Li Y, A review of a class of emerging contaminants: the classification, distribution, intensity of consumption, synthesis routes, environmental effects and

expectation of pollution abatement to organophosphate flame retardants (OPFRs), *Int. J. Mol. Sci.* 20 (12) (2019) 2874.

- [21]. Mills W, A Tataru R, Potential for reuse of e-plastics through processing by compression molding, *Challenges* 7 (1) (2016) 13.
- [22]. Chen M, Ma X, Chen B, Arsenault R, Karlson P, Simon N, Wang Y, Recycling end-of-life electric vehicle lithium-ion batteries, *Joule* 3 (11) (2019) 2622–2646.
- [23]. Alcántara-Concepción V, Gavilán-García A, Gavilán-García IC, Environmental impacts at the end of life of computers and their management alternatives in México, *J. Clin. Prod.* 131 (2016) 615–628.
- [24]. Jin L, et al., Ambient Air Pollution and Congenital Heart Defects in Lanzhou, China. 10(7) 2015, 074005.
- [25]. McAllister L, The human and environmental effects of e-waste, *Popul. Ref. Bur.* (2013).
- [26]. Sahle-Demessie E, et al., Characterizing of Emissions from Open Burning of Electronic Waste using TG-GC-MS System. 2017.
- [27]. Cesaro A, et al., A Relative Risk Assessment of the Open Burning of WEEE. 26(11) 2019, 11042–11052.
- [28]. Li T-Y, et al., Characteristics of Polybrominated Diphenyl Ethers Released From Thermal Treatment and Open Burning of E-waste. 52(8) 2018, 4650–4657.
- [29]. Gullett BK, et al., Characterization of Air Emissions and Residual Ash From Open Burning of Electronic Wastes During Simulated Rudimentary Recycling Operations. 9(1) 2007, 69–79.
- [30]. Needhidasan S, et al., Electronic Waste—An Emerging Threat to the Environment of Urban India. 12(1) 2014, 36.
- [31]. Xie W, Pan W-P, Thermal characterization of materials using evolved gas analysis, *J. Therm. Anal. Calorim.* 65 (3) (2001) 669–685.
- [32]. Gil MV, et al., Thermal Behaviour and Kinetics of Coal/biomass Blends During Co-combustion. 101(14) 2010, 5601–5608.
- [33]. Holland B, Hay J, The thermal degradation of poly (vinyl alcohol), *Polymer* 42 (16) (2001) 6775–6783.
- [34]. Jellinek H, Luh MD, Thermal degradation of polymethylmethacrylate. Energies of activation, *Macromol. Chem. Phys.* 115 (1) (1968) 89–102.
- [35]. Evangelopoulos P, et al., Investigation of the Thermal Decomposition of Printed Circuit Boards (PCBs) Via Thermogravimetric Analysis (TGA) and Analytical Pyrolysis (Py–GC/MS). 115, 2015, 337–343.
- [36]. Vouvoudi EC, et al., Thermal Degradation Characteristics and Products Obtained After Pyrolysis of Specific Polymers Found in Waste Electrical and Electronic Equipment. 11(5) 2017, 9.
- [37]. Wang W, Wu Y, An overview of recycling and treatment of spent LiFePO₄ batteries in China, *Resour. Conserv. Recycl.* 127 (2017) 233–243.
- [38]. Proto AR, Zimbalatti G, Negri M, The measurement and distribution of wood dust, *J. Agric. Eng.* 41 (1) (2010) 25–31.
- [39]. Sezdi M, Investigation of the Performance of the Air Purifier in High-Risk Hospital Rooms by Counting Particles.
- [40]. Guo Q, et al., Pyrolysis of Scrap Printed Circuit Board Plastic Particles in a Fluidized Bed. 198(3) 2010, 422–428.
- [41]. EPA, U., The Toxicity Characteristic Leaching Procedure (TCLP). 1992.
- [42]. USEPA, Methods for Derivation of Inhalation Reference Concentrations and Application of Inhalation Dosimetry (U.E.P.A.), US EPA, Washington, DC, 1994.
- [43]. ASTM. E1131 –08: Standard Test Method for Compositional Analysis by Thermogravimetry. 2014. <https://www.astm.org/READINGLIBRARY/>.
- [44]. Apaydin-Varol E, Polat S, Putun AE, Pyrolysis kinetics and thermal decomposition behavior of polycarbonate—a TGA-FTIR study, *Therm. Sci.* 18 (3) (2014).
- [45]. Xie Q, Zhang H, Tong L, Experimental study on the fire protection properties of PVC sheath for old and new cables, *J. Hazard. Mater.* 179 (1–3) (2010) 373–381. [PubMed: 20359822]

- [46]. Zhu H, et al., TG-FTIR Analysis of PVC Thermal Degradation and HCl Removal. 82 (1) 2008, 1–9.
- [47]. Wang Z, et al., Pyrolysis and Combustion of Polyvinyl Chloride (PVC) Sheath for New and Aged Cables Via Thermogravimetric analysis-Fourier Transform Infrared (TG-FTIR) and Calorimeter. 11(10) 2018, 1997.
- [48]. Conesa JA, et al., Thermogravimetric Studies on the Thermal Decomposition of Polyethylene. 36(1) 1996, 1–15.
- [49]. Dümichen E, et al., Analysis of Polyethylene Microplastics in Environmental Samples, Using a Thermal Decomposition Method. 85, 2015, 451–457.
- [50]. Bizzo W, Figueiredo R, de Andrade V, Characterization of printed circuit boards for metal and energy recovery after milling and mechanical separation, Materials 7 (6) (2014) 4555–4566. [PubMed: 28788692]
- [51]. Heikkinen J, et al., Thermogravimetry as a Tool to Classify Waste Components to be Used for Energy Generation. 71(2) 2004, 883–900.
- [52]. Demirbas A, Pyrolysis of municipal plastic wastes for recovery of gasoline-range hydrocarbons, J. Anal. Appl. Pyrolysis 72 (1) (2004) 97–102.
- [53]. Li K, et al., Investigation into the Catalytic Activity of Microporous and Mesoporous Catalysts in the Pyrolysis of Waste Polyethylene and Polypropylene Mixture. 9(6) 2016, 431.
- [54]. Jung MR, et al., Validation of ATR FT-IR to Identify Polymers of Plastic Marine Debris, Including those Ingested by Marine Organisms. 127, 2018, 704–716.
- [55]. Smidt E, et al., The Application of FT-IR Spectroscopy in Waste Management. 2011, 405–430.
- [56]. Li J, et al., Characteristic of Low-temperature Pyrolysis of Printed Circuit Boards Subjected to Various Atmosphere. 54(11) 2010, 810–815.
- [57]. Evangelopoulos P, Pyrolysis of Waste Electrical and Electric Equipment (WEEE) for Energy Production and Material Recovery. 2014.
- [58]. Zhou Y, Qiu K, A new technology for recycling materials from waste printed circuit boards, J. Hazard. Mater. 175 (1–3) (2010) 823–828. [PubMed: 19939558]
- [59]. Morelli JJ, Thermal analysis using mass spectrometry: a review, J. Anal. Appl. Pyrolysis 18 (1) (1990) 1–18.
- [60]. Wu J, et al., TG/FTIR Analysis on Co-pyrolysis Behavior of PE, PVC and PS. 34(3), 2014, 676–682.
- [61]. Jafari AJ, Donaldson JD, Determination of HCl and VOC emission from thermal degradation of PVC in the absence and presence of copper, copper (II) oxide and copper (II) chloride, J. Chem. 6 (3) (2009) 685–692.
- [62]. Vann K, Musson S, Townsend T, Evaluation of a modified TCLP methodology for RCRA toxicity characterization of computer CPUs, J. Hazard. Mater. 129 (1–3) (2006) 101–109. [PubMed: 16159695]
- [63]. Jang Y-C, Townsend TG, Leaching of lead from computer printed wire boards and cathode ray tubes by municipal solid waste landfill leachates. Environ. Sci. Technol. 37 (20) (2003) 4778–4784.
- [64]. He K, et al., Comparison of Soil Heavy Metal Pollution Caused by E-waste Recycling Activities and Traditional Industrial Operations. 24(10) 2017, 9387–9398.
- [65]. Liu F, et al., Methanolysis of Polycarbonate Catalysed by Ionic Liquid [Bmim][Ac]. 189(1–2) 2011, 249–254.
- [66]. Weeden GS Jr, et al., Method for Efficient Recovery of High-purity Polycarbonates from Electronic Waste. 49(4) 2015, 2425–2433.
- [67]. Cui J, Zhang L, Metallurgical recovery of metals from electronic waste: a review, J. Hazard. Mater. 158 (2–3) (2008) 228–256. [PubMed: 18359555]
- [68]. Gangwar C, et al., Assessment of Air Pollution Caused by Illegal E-waste Burning to Evaluate the Human Health Risk. 125, 2019, 191–199.
- [69]. Zartarian VG, Ozkaynak H, Burke JM, Zufall MJ, Rigas ML, Furtaw EJ, A modeling framework for estimating children's residential exposure and dose to chlorpyrifos via dermal residue contact and nondietary ingestion, Environ. Health Perspect. 108 (6) (2000) 505–514.

- [70]. Agency, U.S.E.P., Risk Assessment Guidance for Superfund; Volume I: Human Health Evaluation Manual (Part A), Environmental Protection Agency Washington DC, 1989.

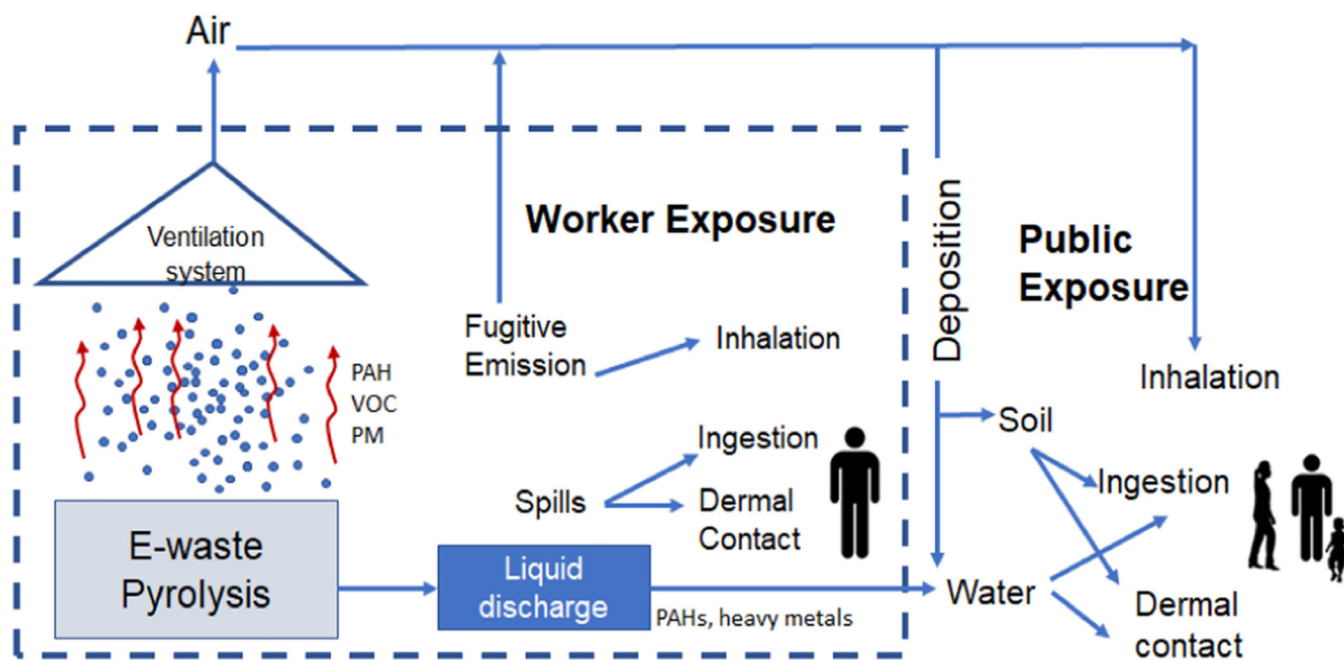


Fig. 1.
Conceptual representation of pollutants emission and exposure from e-waste pyrolysis.

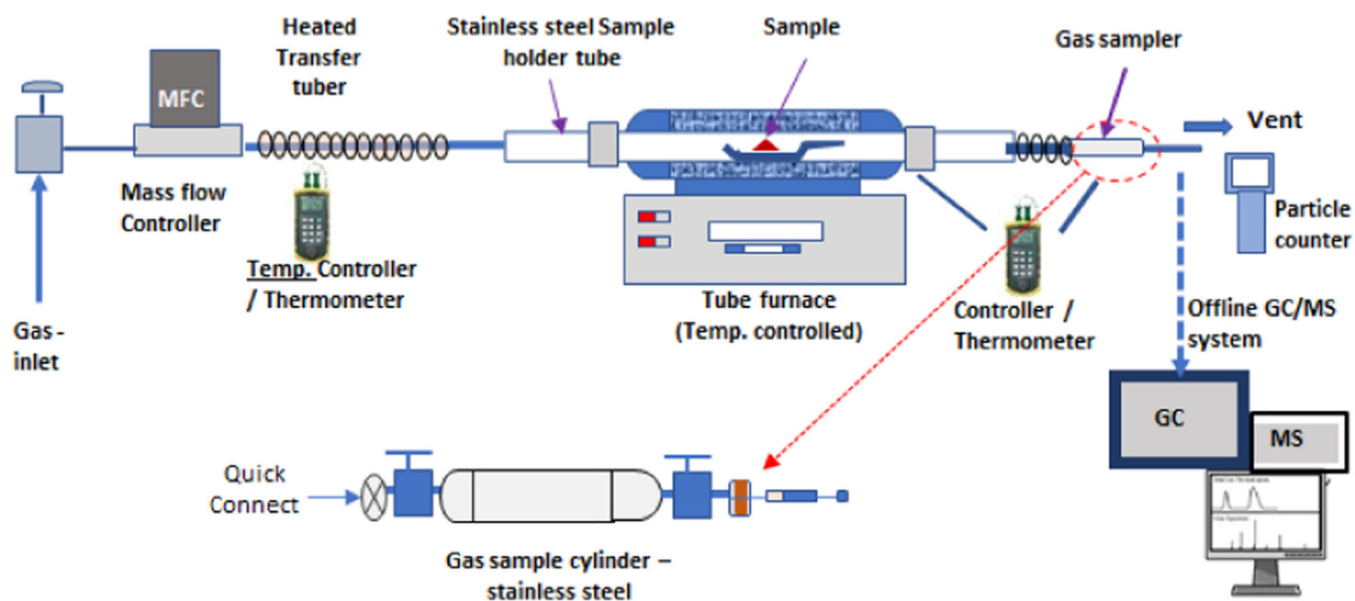


Fig. 2.
Schematic diagram of experimental system for E-waste pyrolysis study.

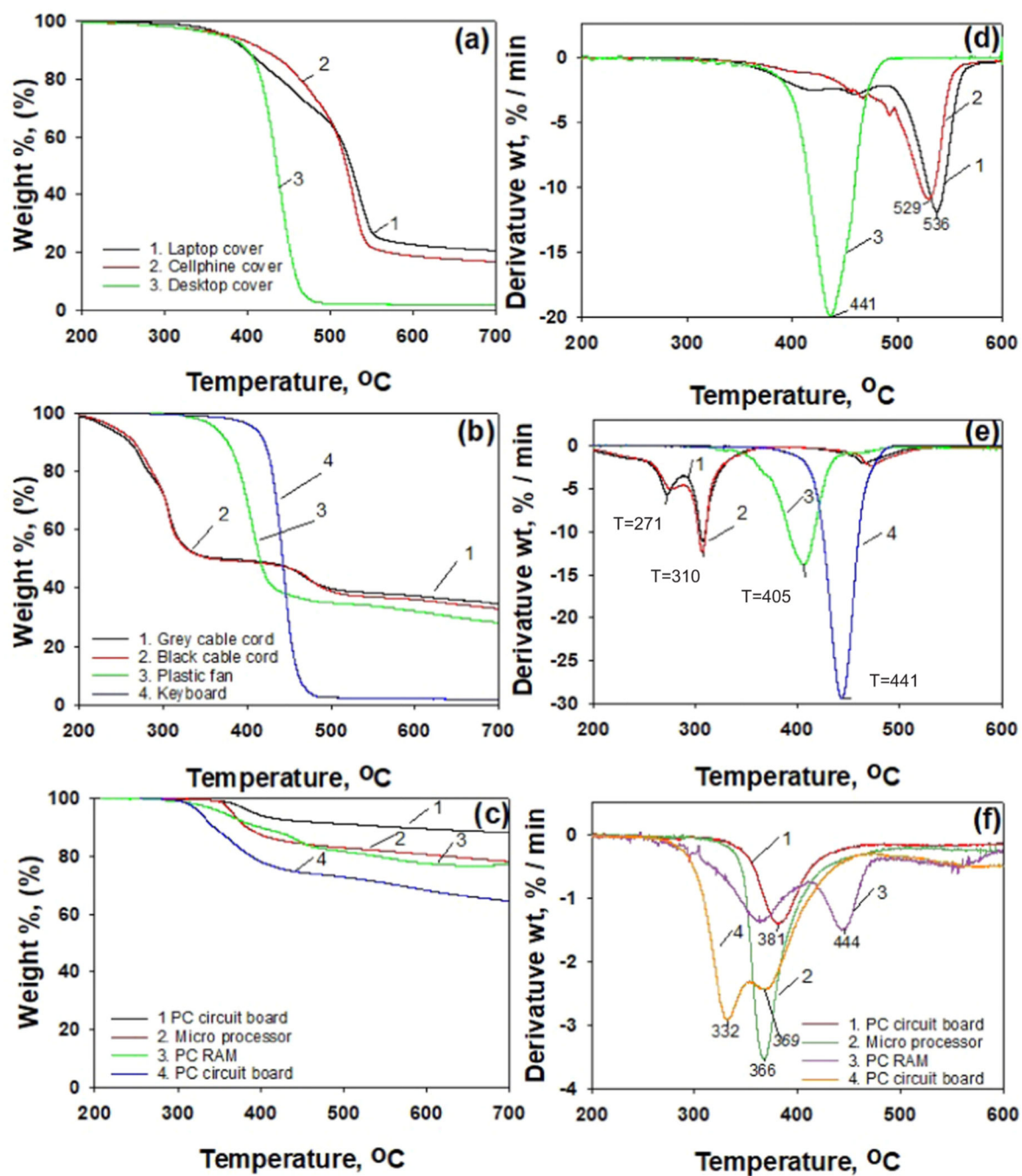


Fig. 3. Thermogravimetric analysis results obtained for e-waste components (a) casings, (b) peripheral, and (c) printed circuit boards; and differential thermal analysis, (d) for casings, (e) for peripheral, and (f) for printed circuit boards.

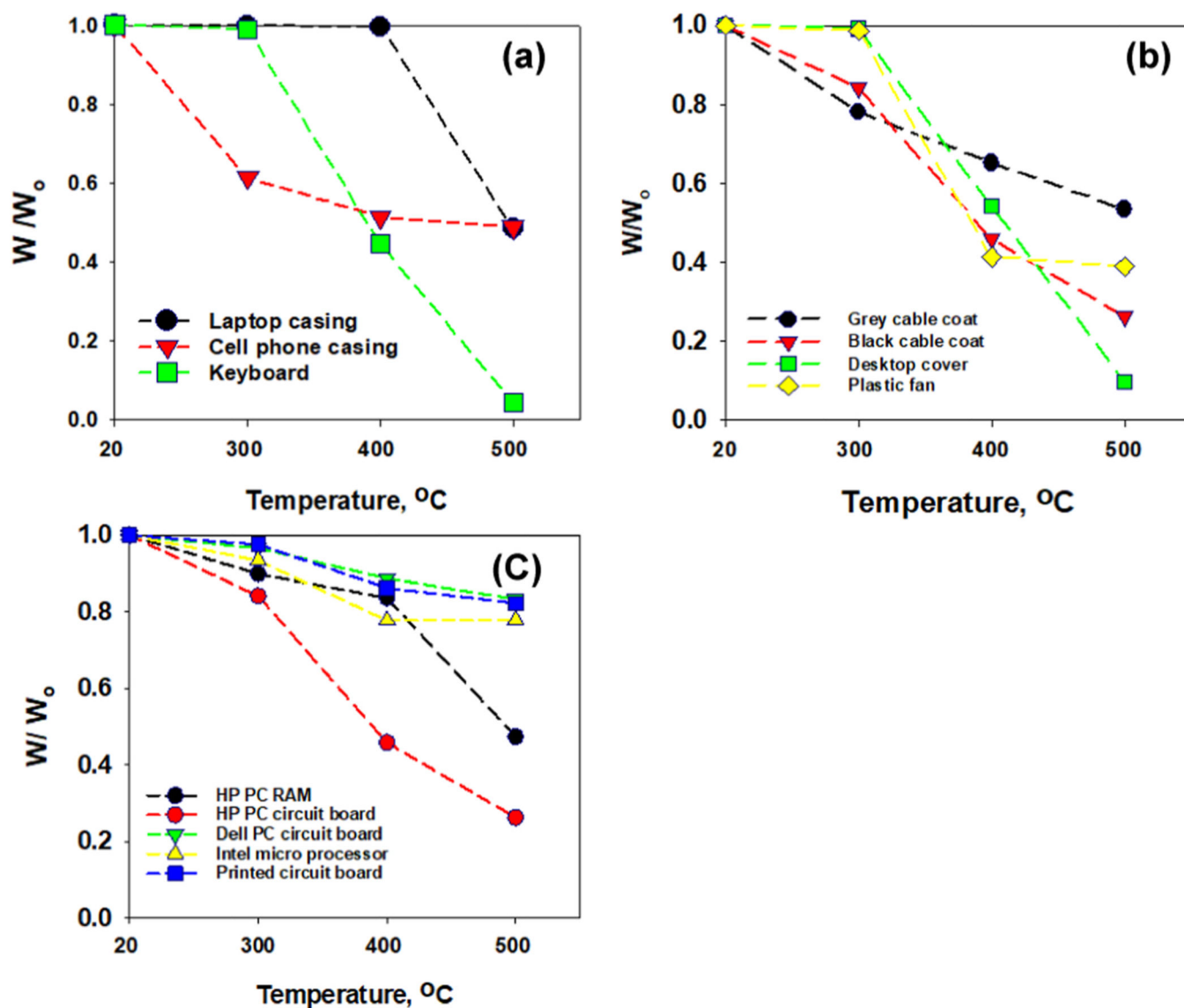


Fig. 4.
Weight loss of e-waste during pyrolysis at selected temperature (a) Casings, (b) peripheral, and (c) printed circuit board.

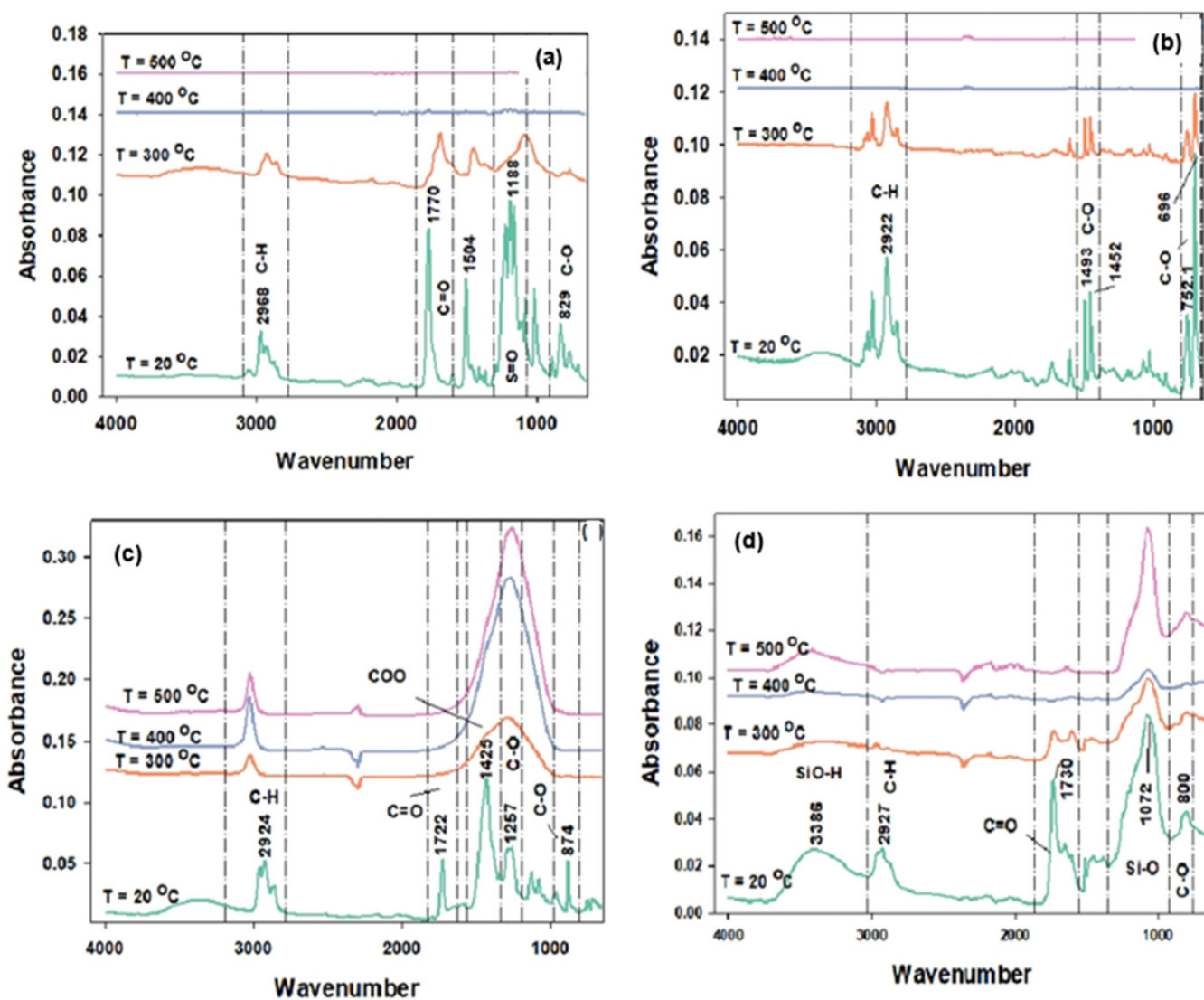


Fig. 5. FTIR spectra of e-waste before and after pyrolysis at selected temperatures for electronic components from (a) cell phone casing, (b) keyboard, (c) plastic cable cord, (d) printed circuit board.

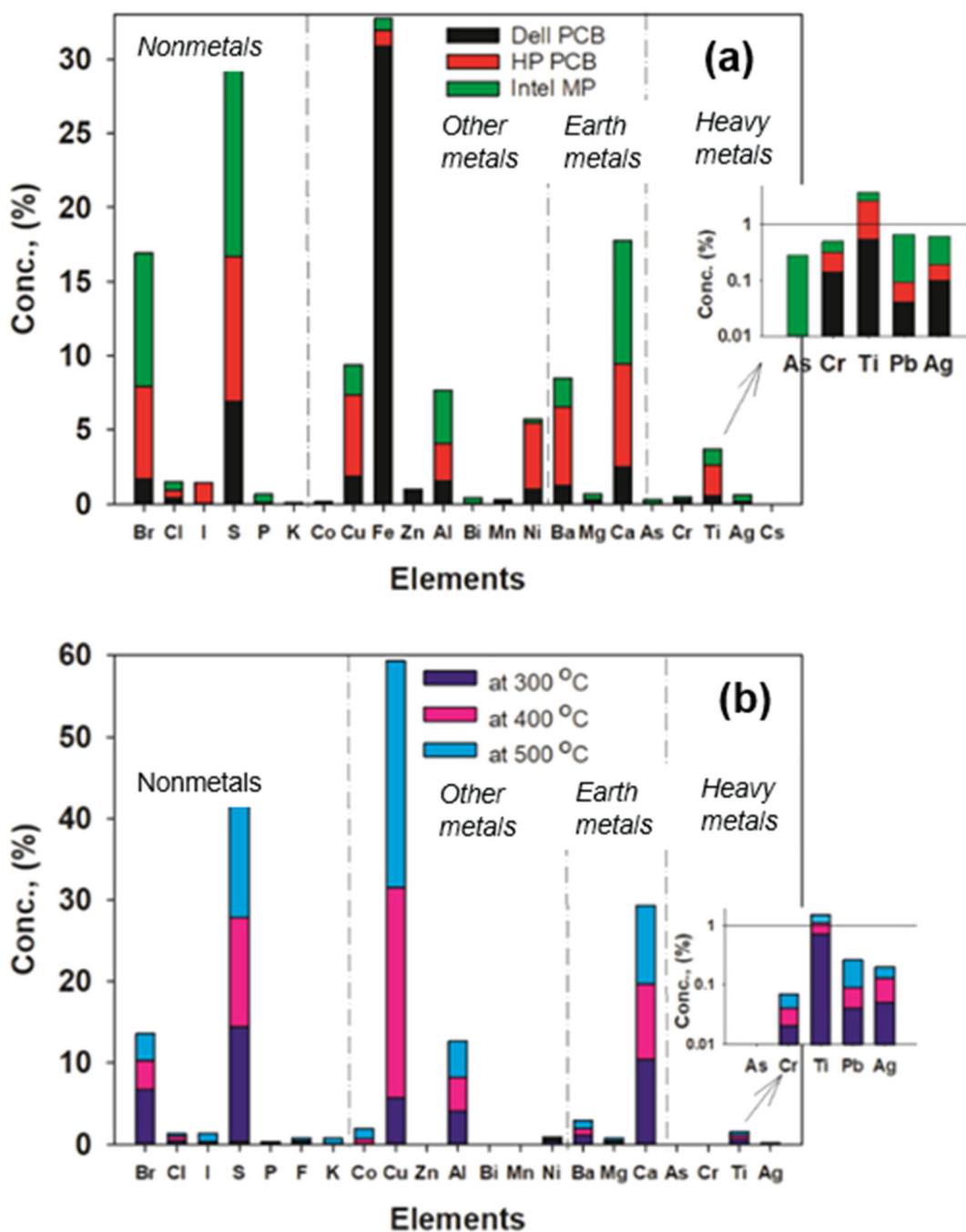


Fig. 6. X-ray fluorescence (XRF) analysis of the chemical composition of showing major elements of (a) printed circuit board from devices different manufactures, (b) effect of pyrolysis temperatures; log scale inserts show the presence of heavy metals at low concentration.

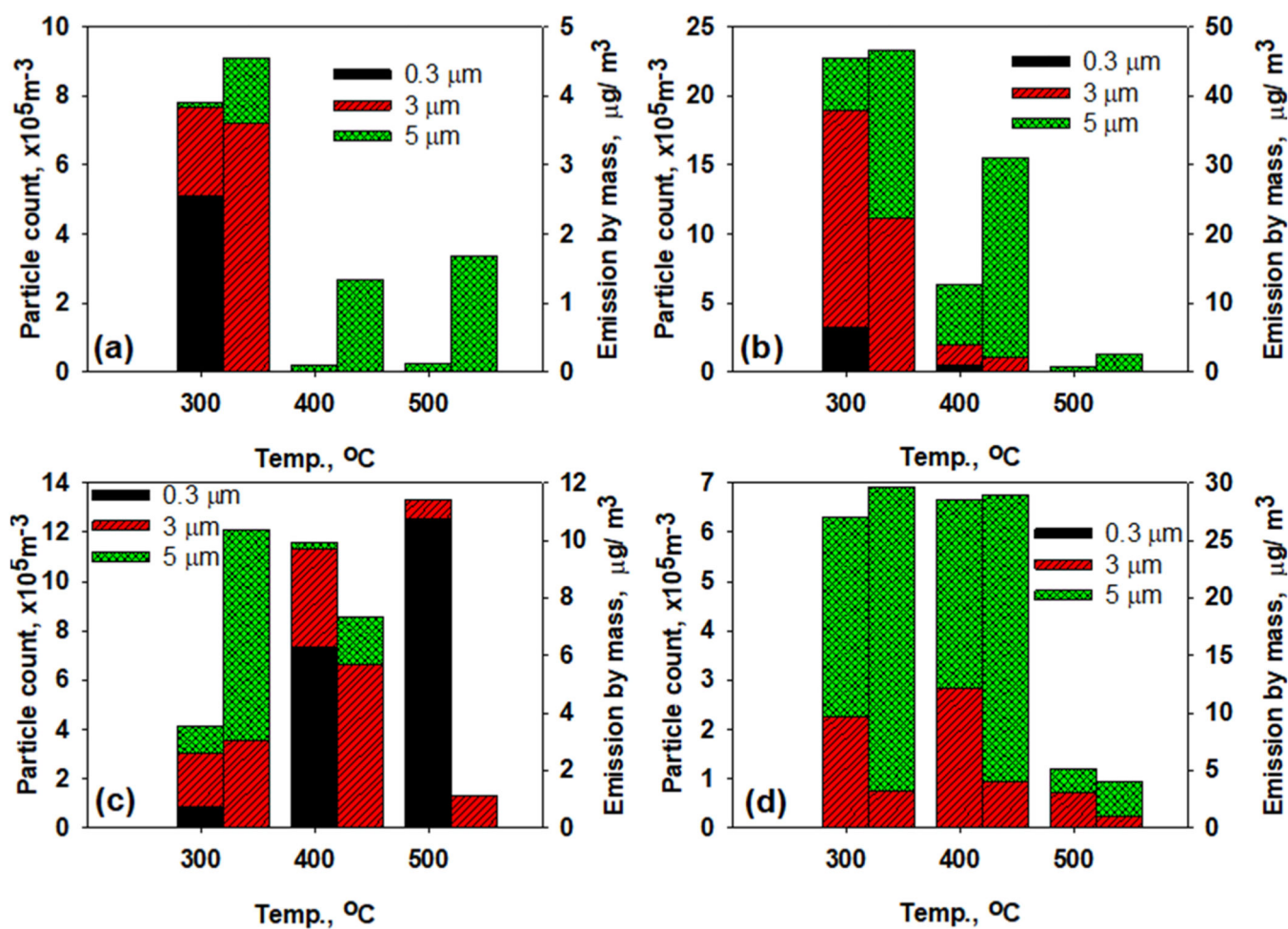


Fig. 7.
Particle emissions from pyrolysis of e-waste at select temperatures (a) electronic plastic casing, (b) plastic desktop cover, (c) plastic cable cord covers, and (d) printed circuit board.

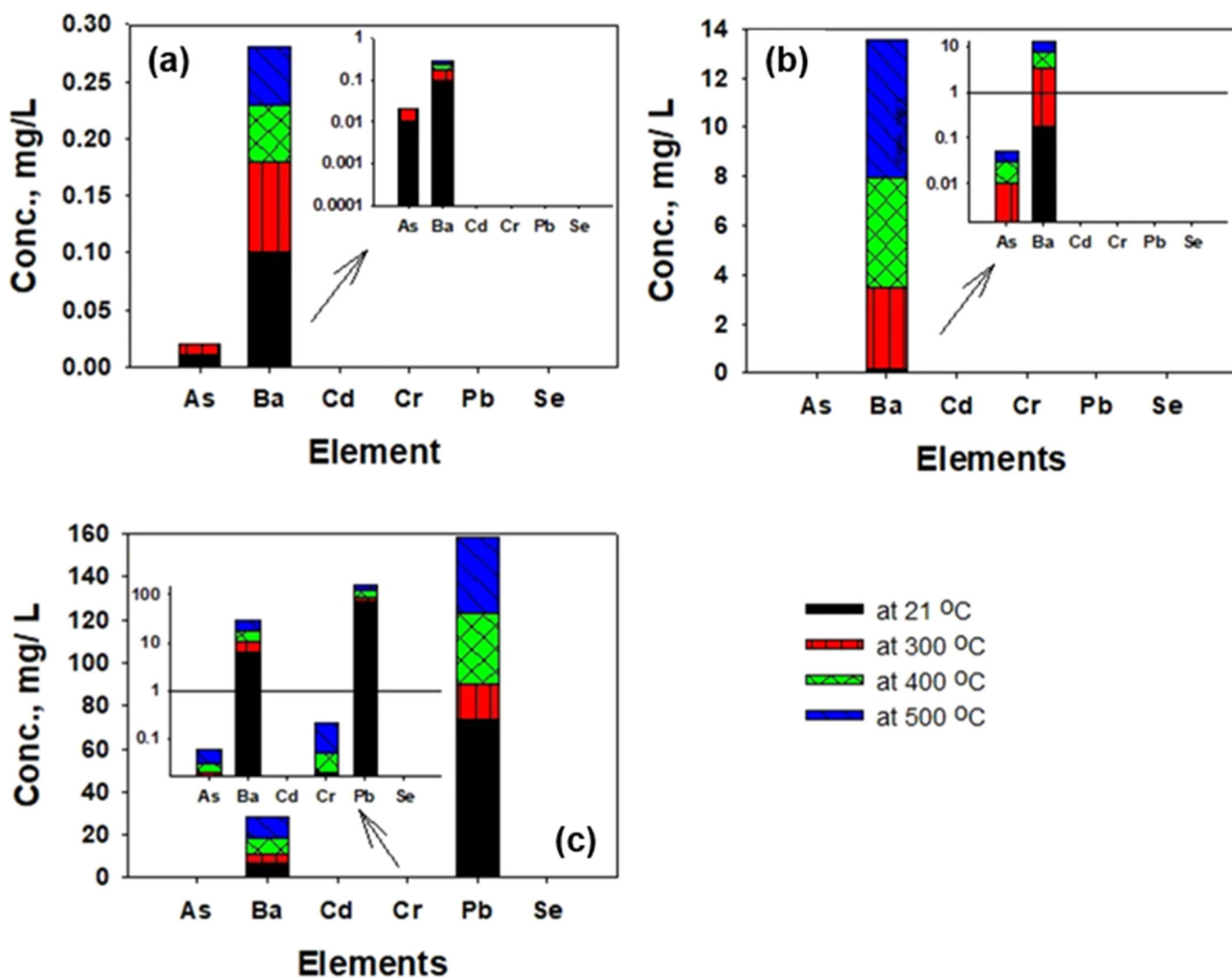


Fig. 8. Comparison of heavy metal leaching from electronics components (a) casings, (b) peripherals, and (c) printed circuit board; log scale inserts show the distribution of leaching metals at lower concentration.

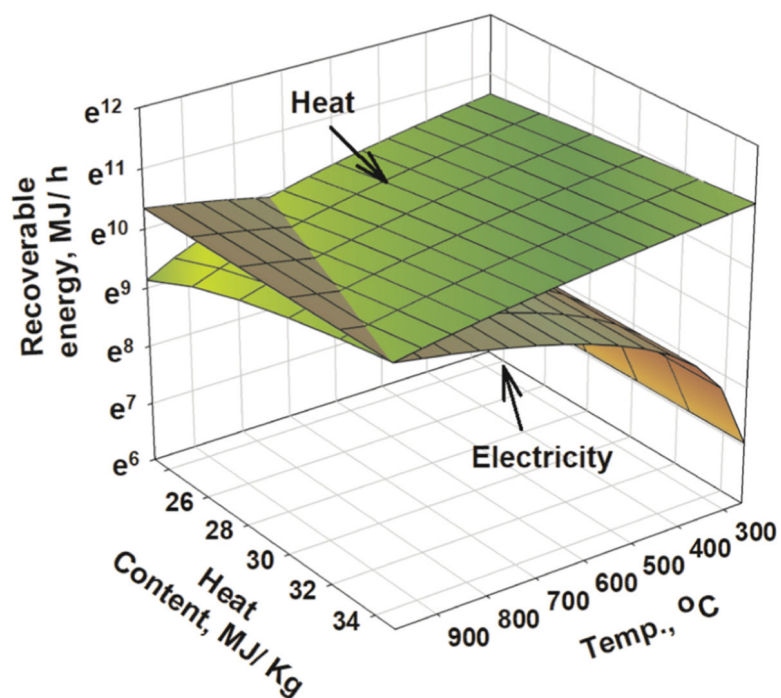


Fig. 9.
Energy recovery from pyrolysis and burning of e-waste.

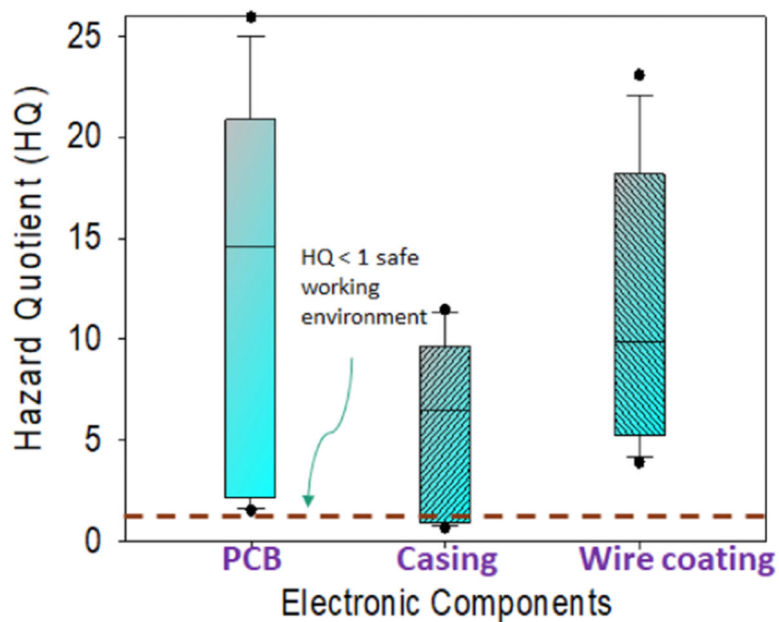


Fig. 10. Hazard quotient of adults chronically exposed to toxic emission of e- aste pyrolysis.

Table 1

Thermogravimetric analysis results on different e-waste components.

Summary of E-waste TGA analysis				
Sample source	Onset temp. (°C)	T ₅₀ (°C)	T _{max} (°C)	Ash remaining (%)
Casings				
Laptop casing	367	530	536	20
Cellphone casing	416	517	529	15
Desktop cover	412	439	441	2
Peripheral				
Gray cable cord	224	309	309	34
Black cable cord	260	311	310	32
Keyboard	422	438	441	1.5
Fan	377	401	405	28
Printed circuit boards				
Dell PCB	366	377	381	87
Intel MP PCB	354	368	366	23
Dell PCB	352	368	366	79
HP RAM PCB	334	440	444	77
HP PCB	323	368	369	64

Table 2

Electronic component materials categories and source.

Category	Common materials	Source
Printed circuit board	Fiberglass, epoxy resin with a copper foil, paper reinforced phenolic resin, copper, lead, nickel, gold, aluminum, polyimide, polyester, Teflon	Dell desktop HP desktop RAM Intel microprocessor
Casings	PC + ABS - FR(40)	Laptop
	PC	Cellphone
	PC + ABS	Desktop cover
Peripheral	PVC	Gray plastic cable
	PVC	Black plastic cable
	PC + ABS	Keyboard
	Polystyrene and acrylate	Plastic fan

PC = polycarbonate, ABS = acrylonitrile butadiene styrene, FR(40) = flame retardant (i.e Tetrabromobisphenol A (TBBPA), triaryl phosphate ester, PVC = polyvinyl chloride, RAM = random access memory.

Table 3

Chemical composition of gas phase products from pyrolysis of electronic components where a – casing, b- electronic peripherals, c- PCB.

Compound	Molec. formula	CAS #	% available	Category	Health risk assessment (IRIS)		
					RfD (mg/kg-day)	RfC (mg/m³)	Carcinogenicity
Aromatics							
Phenol	C ₆ H ₅ OH	108-95-2		a, c	3 × 10 ⁻¹	N/A	☛
Benzene	C ₆ H ₆	71-43-2	19-71	a, c	4 × 10 ⁻³	3 × 10 ⁻²	☛
Styrene	C ₈ H ₈	100-42-5	5-46	a, b	2 × 10 ⁻¹	1	§
Toluene	C ₇ H ₈	108-88-3	21-72	a, b, c	8 × 10 ⁻²	5	☛
Benzene, 1,3 dimethyl	C ₈ H ₁₀	108-38-3	25-44	a, b	N/A	N/A	**
Ethylbenzene	C ₈ H ₁₀	100-41-4	8-51	a	1 × 10 ⁻¹	1	§
α-Methylstyrene	C ₉ H ₁₀	98-83-9	29-58	a, b	5 × 10 ⁻²	N/A	§
P-cresol	C ₇ H ₈ O	106-44-5	6-11	c	N/A	N/A	☛
p-Xylene	C ₈ H ₁₀	106-42-3	6-9	b	2 × 10 ⁻¹	1 × 10 ⁻¹	☛
o-Xylene	C ₈ H ₁₀	95-47-6	5-9	b	2 × 10 ⁻¹	1 × 10 ⁻¹	☛
Acetophenone	C ₈ H ₁₀ O	98-86-2	25-60	a, b	N/A	N/A	*
Butanedioic acid, phenyl	C ₁₀ H ₁₀ O ₄	635-51-8	0- ⁻²⁰	a, b	N/A	N/A	*
Oxazole	C ₃ H ₃ NO	288-42-6	0-57	c	N/A	N/A	*
phosphonic acid,	C ₆ H ₇ O ₄ P	33795-18-5	12-19	c	N/A	N/A	*
Benzaldehyde	C ₇ H ₆ O	100-52-7	10-73	a	1 × 10 ⁻¹	N/A	☛
Benzeneacetamide	C ₈ H ₉ NO	103-81-1	0-66	a	N/A	N/A	*
2-Vinylfuran	C ₆ H ₆ O	1487-18-9	0-28	a	N/A	N/A	☛
Benzo furan , 2-methyl	C ₉ H ₈ O	4265-25-2	36-43	c	N/A	N/A	*
Aliphatic hydrocarbons							
1-Hexene, 3,5 dimethyl	C ₈ H ₁₆	7423-69-0	12-47	b	N/A	N/A	*
Chloroundecane	C ₁₁ H ₂₃ Cl	2473-03-2	0-13	b	N/A	N/A	*
2-Propenenitrile	C ₃ H ₃ N	107-13-1	0-95	b	N/A	N/A	☛
butane, 1-chloro-3, 3-dimethyl	C ₆ H ₁₃ Cl	2855-08-5	0-13	b	N/A	N/A	*
2-Propenal, 3-phenyl	C ₉ H ₈ O	104-55-2	0-21	b	N/A	N/A	*

Compound	Molec. formula	CAS #	% available	Category	Health risk assessment (IRIS)		
					RfD (mg/kg-day)	RfC (mg/m ³)	Carcinogenicity
1,3,5,7cyclooctatetraene	C ₈ H ₈	629–20-9	25–29	a, b, c	N/A	N/A	*
Bromine containing compounds							
Phenol, 2-bromo	C ₆ H ₅ BrO	95–56-7	41–46	a, c	N/A	N/A	⚡
Phenol, 3-bromo	C ₆ H ₅ BrO	591–20-8	13–24	a, c	N/A	N/A	⚡
Benzene,bromo	C ₆ H ₅ Br	108–86-1	0–98	c	N/A	N/A	*
Isopropenyl bromide	C ₃ H ₅ Br	557–93-7	0–29	c	N/A	N/A	*
1-Propane, 1 bromo	C ₃ H ₇ Br	106–94-5	0–49	c	N/A	N/A	⚡
Bromoacetone	C ₃ H ₅ BrO	598–31-2	0–96	c	N/A	N/A	⚡
Benzyl bromide	C ₇ H ₇ Br	100–39-0	0–22	a	N/A	N/A	⚡
Pantane, 2 = bromo-2-methyl	C ₆ H ₁₃ Br	4283–80-1	8–12	c	N/A	N/A	*

⚡= possibly human carcinogen, IRIS
§= possibly human carcinogen, IARC
☛= exposure can cause organ damage, CDC
⚡*= may be harmful if exposed (TOXNET)
*= no adequate information

Table 4

Chemical composition of liquid phase products from pyrolysis of electronic.

Compound	Formula	CAS #	% available	Category	Health risk assessment		
					RfD (mg/kg-day)	RfC (mg/m³)	Carcinogenicity
Aromatic compounds							
Styrene	C ₈ H ₈	100-42-5	4.1	a	2 × 10 ⁻¹	1	§
α-Methylstyrene	C ₉ H ₁₀	98-83-9	2.43	a	N/A	N/A	♣
p-Crestol	C ₇ H ₈ O	106-44-5	6.05-8.29	a	N/A	N/A	♣
Terphenyl	C ₁₈ H ₁₄	612-71-5	3.47	a	N/A	N/A	§
Phenol	C ₅ H ₅ OH	108-95-2	1.92-59.49	a, c	3 × 10 ⁻¹	N/A	♣
2-Isopropylphenol	C ₉ H ₁₂ O	88-69-7	16.26	a	N/A	N/A	*
Bisphenol A	C ₁₅ H ₁₆ O ₂	80-05-7	12.02-40.49	a, c	5 × 10 ⁻²	N/A	*
4-Phenylphenol	C ₁₂ H ₁₀ O	92-69-3	9.65-26.82	c	N/A	N/A	✱
4-Tert-Butylphenol	C ₁₀ H ₁₄ O	98-54-4	9.65	c	N/A	N/A	✱
Phthalic anhydride	C ₈ H ₄ O ₃	85-44-9	9.53-80.68	c	2	N/A	♣
Benzoic acid	C ₇ H ₆ O ₂	65-85-0	33.82	a	4	N/A	*
4-Hydroxybenzoic acid	C ₇ H ₆ O ₃	99-96-7	4.17	a	N/A	N/A	*
Methyl palmitate	C ₁₇ H ₃₄ O ₂	112-39-0	5.31	b	N/A	N/A	*
Dimethyl phthalate	C ₁₀ H ₁₀ O ₄	131-11-3	7.68-11.02	b	N/A	N/A	♣
1,3-Diphenylpropane	C ₁₅ H ₁₆	1081-75-0	31.16	a	N/A	N/A	♣
4-Phenylbutyronitrile	C ₁₀ H ₁₁ N	2046-18-6	25.63	a	N/A	N/A	♣
Bromine containing compounds							
2-Bromophenol	C ₆ H ₅ BrO	95-56-7	4	c	N/A	N/A	✱
4-Bromophenol	C ₆ H ₅ BrO	106-41-2	4	c	N/A	N/A	*
2,6-Dibromophenol	C ₆ H ₄ Br ₂ O	608-33-3	4.54	c	N/A	N/A	*

♣= possibly human carcinogen, IIRIS

§= possibly human carcinogen, IARC

♣= exposure can cause organ damage, CDC.

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 =
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